Athletic Readiness: Data Generation

Arpit Gajera – AU2340051
Harsh Vipul Patel – AU2340083
Vatsalkumar Parsaniya – AU2340145
Keval Gajjar – AU2340220
Meetkumar Tandel – AU2340230
Team Name: Deep Vision
Department of Computer Science and Engineering
Course: CSE 641 Computer Vision

***Abstract*—This report presents the progress of a computer vision project focused on generating privacy-preserving synthetic athlete performance data. Over the first four weeks, the project progressed through problem formulation, data collection via web scraping, data preprocessing and exploratory data analysis (EDA) and initial exploration of generative modeling techniques. A structured dataset of NBA player statistics was created, cleaned, and normalized to support machine learning workflows. Additionally, Variational Autoencoders (VAE) and Generative Adversarial Networks (GANs) were studied to guide the next phase of synthetic data generation. The results establish a strong foundation for building generative models that preserve statistical properties while ensuring data privacy.**

***Index Terms*—Synthetic Data, Generative Models, VAE, GAN, Data Preprocessing, EDA, Computer Vision**

I. Introduction

In sports analytics, athlete performance data is often sensitive and cannot be openly shared. The goal of this project is to generate realistic synthetic data that preserves statistical characteristics without exposing real identities.

During Week 1, the problem was defined and relevant datasets were identified. The focus was on understanding athlete-level performance metrics and planning a data collection strategy using web scraping tools such as BeautifulSoup and Selenium, but we were able to successfully web scrape the data without using any tools.

In Week 2, a structured dataset was created by scraping NBA player statistics from ESPN, resulting in 500 records with 19 performance features. Later, we got to know that the data generated was not sufficient in order to perform VAE and GANs after which the dataset was updated to 1322 rows.

Week 3 focused on preprocessing and exploratory data analysis (EDA), transforming the dataset into a clean and normalized format suitable for machine learning .

In Week 4, the project transitioned toward generative modeling by studying VAE and GAN architectures, which will be used for synthetic data generation in subsequent phases.

II. Methodology

A. Data Collection

Data was collected through web scraping from ESPN's NBA statistics pages. The dataset includes:

- 5 seasons × 2 subseasons (regular + postseason)
- 130 players per subseason approx
- Total: 1322 records

Features include scoring, rebounds, assists and defensive metrics.

B. Data Preprocessing

Data preprocessing steps included:

- Removal of irrelevant columns (i.e. rank)
- Handling missing values (none found after cleaning)
- Conversion of string features to numeric types
- Encoding categorical variables (player positions)
- Outlier detection using IQR method

C. Exploratory Data Analysis

EDA involved:

- Statistical summaries (mean, std, CV)
- Distribution analysis (histograms, skewness)
- Correlation analysis (heatmaps)

TABLE I
Feature Selection and Dimensionality Reduction

Feature	Action	Rationale (r = corr. with PTS)
NAME	Drop	Identifier; non-numeric.
POS	Drop	Handled separately by POS_ENC.
GP	Drop	Low correlation ($r = 0.31$); reflects availability.
MIN	Keep	High correlation ($r = 0.88$); workload proxy.
PTS	Keep	Primary target metric.
FGM	Drop	Collinear with FGA ($r = 0.97$).
FGA	Keep	Shot volume metric ($r = 0.98$).
FG%	Keep	Independent efficiency metric ($r = 0.28$).
3PM	Drop	Collinear with 3PA ($r = 0.95$).
3PA	Keep	3PT volume metric ($r = 0.72$).
3P%	Keep	Independent efficiency metric ($r = 0.30$).
FTM	Drop	Collinear with FTA ($r = 0.98$).
FTA	Keep	Foul-drawing volume ($r = 0.87$).
FT%	Keep	Independent efficiency metric ($r = 0.50$).
REB	Keep	Key performance metric ($r = 0.71$).
AST	Keep	Key performance metric ($r = 0.79$).
STL	Keep	Key defensive metric ($r = 0.66$).
BLK	Keep	Key defensive metric ($r = 0.43$).
TO	Keep	Usage and ball-handling metric ($r = 0.83$).
DD2	Drop	Zero-inflated (77.2% zeros).
TD3	Drop	Zero-inflated (97.5% zeros).
SEASON	Drop	No correlation ($r = -0.004$); metadata.

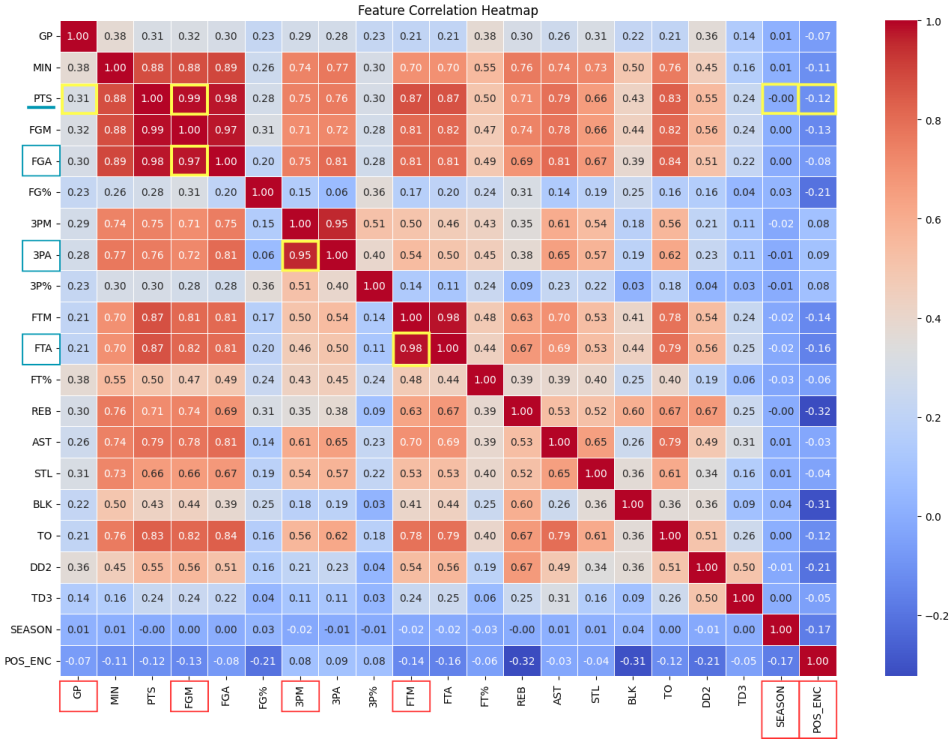

Fig. 1. Feature Correlation Heatmap of NBA player statistics. Highlighted regions indicate features with near-perfect collinearity (e.g., FGM and FGA, r=0.97) or negligible correlation to target performance metrics, which were subsequently removed to optimize the generative models.

Findings showed strong correlations between scoring metrics and playing time, and right-skewed distributions across most features.

D. Feature Scaling

Standardization (zero mean, unit variance) was applied to improve model training stability and the dataset was produced.

E. Generative Model Exploration

In Week 4, we explored:

- Variational Autoencoders (VAE): Understanding encoder-decoder structure and latent space representation
- Generative Adversarial Networks (GANs): Studying generator-discriminator dynamics

This exploration will guide model selection for synthetic data generation.

III. Results

- Successfully created a structured dataset of NBA player statistics
- Identified strong correlations between scoring and playing time
- Observed right-skewed distributions across most features
- Prepared normalized dataset for generative modeling

IV. Discussions

The preprocessing pipeline ensured high-quality data suitable for machine learning. The presence of skewed distributions and strong correlations highlights the importance of generative models that can capture complex relationships.

Outlier removal improved dataset consistency but reduced sample size, which may affect model generalization. Additionally, class imbalance in player positions may influence synthetic data quality.

The study of VAE and GAN architectures in Week 4 provides critical insights into how latent representations and adversarial training can be leveraged to generate realistic synthetic data.

V. Conclusion

The first four weeks of the project successfully established a strong foundation for synthetic data generation. Data collection, preprocessing, and EDA were completed, and the dataset is now ready for modeling.

The exploration of VAE and GAN models in Week 4 marks the transition from data preparation to model development. Future work will involve implementing these models to generate high-quality synthetic athlete performance data while preserving statistical integrity.

References

[1] ESPN NBA Player Statistics Datasethttps://www.espn.in/nba/stats/player
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[3] Goodfellow, I. et al. (2014). Generative Adversarial Networks